

# Gradient-Sensitive Functional Descriptors for Interferometric Surface Metrology: Complementing Amplitude-Averaging ISO Parameters

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## Abstract

This paper presents a complete interferometric pipeline for optical evaluation of roundness and surface profiles using a self-built Michelson interferometer. The method is validated on two certified JENOPTIK standards: the roundness standard FN 111 (24,000 interferograms) and the magnification standard FN 101 with a certified Flick flat (100,000 interferograms), each measured in five  $400^\circ$  sweeps. External validation against 104 tactile reference measurements yields an  $E_n$  score of 0.61 for the FN 111 and 0.3% agreement for the FN 101, demonstrating metrological consistency of the optical reconstruction. In addition to conventional profile roughness parameters (ISO 21920-2, formerly ISO 4287) and roundness metrics ( $RONt$ ,  $RONq$ ), the reconstructed nanometre profiles are analysed using functional-analytic descriptors including  $L^p$ , Sobolev  $W^{1,2}$ , and bounded-variation (BV) norms. Controlled defect-injection experiments and analysis of the certified Flick flat show that gradient-based descriptors provide complementary sensitivity to localised geometric features that may remain below the detection threshold of amplitude-averaging parameters. For a 100 nm synthetic spike, BV density increased by 18% and the  $W^{1,2}$  seminorm by 622%, while  $R_a$  showed no detectable response at the reconstruction resolution. The BV descriptor also detected the certified Flick flat in a sliding-window analysis where  $R_a$  remained insensitive. These results suggest that functional descriptors can complement standard profile and roundness parameters, providing additional information on localised geometric variations without requiring changes to the measurement hardware. All reported uncertainties are within-

run repeatability estimates from five sweeps of a single continuous rotation and should be interpreted as lower bounds; an indicative type-B uncertainty budget is provided.

*Keywords:* interferometry, roundness, surface profile, functional norms, bounded variation, Sobolev seminorm, uncertainty

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## 1. Introduction

Surface topography is one of the most important properties of technical parts. The accuracy of form and roughness influences contact conditions, wear and lifetime. Roundness, understood as deviation from a perfect circle, remains a central requirement in precision manufacturing; Zhu et al. emphasised that precise measurements are needed both for form tolerances and for tribological surface texture assessment [1]. At nanometric scale, such measurements require high-precision optical systems [2]. The foundational reference for surface metrology terminology and parameter definitions is the handbook of Whitehouse [3].

Optical dimensional metrology has a long connection with interference [4]. Modern optical surface metrology relies on diffraction, polarisation and interference [5, 6]. Diffraction and scattering have been applied to roughness measurement in several distinct ways: Brodmann introduced a scattered-light roughness instrument for production environments [7] and subsequently extended the scattered-light principle to combined roughness, form and waviness measurement [8]; Fan and Huynh analysed light scattering from flat periodic rough surfaces to estimate roughness parameters [9]; Hertzsch et al. demonstrated model-based scatterometry for microtopographic analysis of turned surfaces [10]; and a low-cost scatterometry setup achieving nanometric precision in fast roughness measurement has been reported previously [11]. Interference remains central for ultra-precise surface-form measurement, from the early interference-based length comparisons of Benoît [12] to modern large-scale precision optics programmes [13]. De Groot systematised the principles of phase-shifting and coherence-scanning interference microscopy for topography evaluation [14, 15],

25 and Chatterjee described phase-evaluation interferometric testing of reflective  
26 surfaces [16].

27 Several interferometric instruments have been proposed for high-accuracy  
28 surface and roundness measurement. Polack et al. described a Michelson-based  
29 system for surface-shape determination built around a high-resolution stitching  
30 interferometer developed by Thomasset et al. [17, 18]. Fan et al. presented  
31 a roundness measuring system for microspheres based on two small Michel-  
32 son interferometers [19]. Kuhnel et al. proposed an interferometric system for  
33 roundness and cylindricity measurement of ring gauges with 0.1 nm resolution  
34 over a large measurement range [20]. Commercial instruments can be highly ac-  
35 curate, but they are often costly and may not expose the full acquisition chain  
36 needed for experimental method development.

37 The numerical analysis of fringe patterns has its own methodological lineage,  
38 which frames the phase-decoding route used here. Takeda et al. introduced the  
39 Fourier-transform method of fringe-pattern analysis, in which a spatial carrier  
40 allows the phase to be recovered from a single interferogram without temporal  
41 phase shifting [21]. Kemaio extended this concept with the two-dimensional win-  
42 dowed Fourier transform, which processes the fringe pattern locally to improve  
43 robustness to noise and local fringe-quality variations [22]. More recently, Feng  
44 et al. demonstrated that deep neural networks can be trained to perform fringe  
45 analysis, substantially enhancing the accuracy of phase demodulation from a  
46 single fringe pattern [23]. Galaktionov and Toporovsky proposed a multi-stage  
47 fringe-map reconstruction algorithm for fibre-end surface analysis in non-phase-  
48 shifting interferometry, combining moving-average and Fourier-transform filter-  
49 ing with polynomial fitting and reporting agreement within 2% of a commercial  
50 phase-shifting interferometer [24]. The radial excess-fraction method used in the  
51 present work belongs to this single-frame, non-phase-shifting family: each inter-  
52 ferogram is decoded independently through the geometry of its circular fringes,  
53 which suits continuous-rotation acquisition where temporal phase stepping is  
54 impractical.

55 The present work uses a self-built, non-contact Michelson-type roundness

56 measurement system assembled from off-the-shelf components. The optical  
 57 setup, apparatus description and radial image-processing route for phase extrac-  
 58 tion were published in a previous study [25], where only roughness parameters  
 59 were reported for the ceramic standard FN 111. The present work extends that  
 60 study in three directions: (i) roundness ( $RONt$ ,  $RONq$ ) is evaluated for the first  
 61 time with this self-built optical setup, (ii) a second artefact — the JENOPTIK  
 62 magnification standard FN 101 with a Flick flat — is measured and validated  
 63 against its DAkkS-DKD certificate, and (iii) a controlled sensitivity-comparison  
 64 methodology is introduced: synthetic defects of known geometry are injected  
 65 into measured profiles, and the response of classical ISO parameters is com-  
 66 pared against that of  $L^p$ , Sobolev  $W^{1,2}$ , and bounded-variation (BV) norms  
 67 — standard tools of functional analysis here applied to interferometric profile  
 68 metrology — all computed from the same nanometre residual. The comparison  
 69 framework, rather than any individual norm, constitutes the methodological  
 70 contribution.

71 The computational problem is that modern interferometric surface metrol-  
 72 ogy records long image sequences before a numerical profile exists. In the present  
 73 data, however, each PNG file is already a separate interferogram in which the  
 74 phase is encoded spatially by the bright and dark fringes inside the image. The  
 75 files in the FN 111 and FN 101 folders are therefore not treated as a temporal  
 76 phase-shifting stack of one fixed scene. Instead, each image  $I_i(x, y)$  is processed  
 77 independently to obtain one wrapped excess-fraction value  $\epsilon_i$ . The ordered se-  
 78 quence of these values is then unwrapped over image order and converted to a  
 79 nanometre profile. The full measurement archive contains 24,000 grayscale in-  
 80 terferograms of the FN 111, with image size  $648 \times 488$  px, and 100,000 grayscale  
 81 interferograms of the JENOPTIK magnification standard FN 101, with image  
 82 size  $640 \times 480$  px.

83 The literature relevant to this comparison is mature but only partly con-  
 84 nected. In interferometric testing, Dumas et al. extended the lateral range of  
 85 interferometry through subaperture stitching [26]; Chen et al. developed an it-  
 86 erative stitching algorithm for spherical surfaces [27] and subsequently applied

87 stitching to the testing of large optical surfaces [28]; Iacchetta et al. addressed  
 88 rotation and translation registration of band-limited interferometric images us-  
 89 ing a chirp z-transform [29]; and Burge et al. applied computer-generated holo-  
 90 grams to the measurement of x-ray and EUV optics [30]. On repeatability,  
 91 Quan et al. compared evaluation approaches for interferometric surface repeata-  
 92 bility [31], and Zhang et al. compared repeatability across different interference  
 93 systems [32]; both characterise repeatability through RMS/PV quantities or  
 94 pixel-wise standard-deviation matrices, i.e., through amplitude statistics. Pap-  
 95 pas et al. propagated the uncertainty of areal surface-texture parameters using  
 96 the metrological-characteristics approach, embedding texture parameters in a  
 97 GUM-style framework [33]. In statistics, Gertheiss et al. reviewed functional  
 98 data analysis as a language for function-valued observations [34], and Happ et  
 99 al. provided a framework separating amplitude and phase variation in functional  
 100 data [35]. Scale-dependence has been examined by Wolf, who connected surface  
 101 topography with topological description [36], by Sanner et al., who defined scale-  
 102 dependent roughness parameters for topography analysis [37], and by Zhao et  
 103 al., who studied fractal simulation of topography for lubrication prediction [38];  
 104 collectively these show that surface characterisation depends on bandwidth and  
 105 filtering scale.

106       None of these strands, however, resolves a specific practical problem of pro-  
 107 file and roundness metrology. When a localised geometric feature — a flat, a pit,  
 108 a scratch — is present on an otherwise smooth artefact, amplitude-averaging  
 109 parameters such as  $R_a$  dilute its signature over the entire evaluation length,  
 110 so the feature can remain numerically invisible although it is metrologically  
 111 relevant. Existing remedies do not fully address this for profile data: areal  
 112 parameters require calibrated height maps that single-image radial extraction  
 113 does not provide; visual inspection of polar plots is operator-dependent and not  
 114 quantitative; and template-based defect detectors require the defect geometry  
 115 to be known in advance. The unresolved question addressed here is therefore  
 116 whether gradient-sensitive functional norms — whose mathematical structure  
 117 makes them respond to localised slope content rather than to averaged am-

118 plitude — provide reproducible, quantifiable sensitivity to such features when  
 119 computed from the same reconstructed residual as the standard ISO param-  
 120 eters, and how large that sensitivity advantage is under controlled conditions.  
 121 What has been missing in the literature is precisely this controlled, like-for-like  
 122 comparison on identical nanometre residuals with external tactile validation.  
 123 The present work addresses this by: (i) injecting synthetic defects of known ge-  
 124 ometry into measured profiles to quantify descriptor response under controlled  
 125 conditions; (ii) applying the full pipeline to two DAkS-DKD-certified arte-  
 126 facts at large scale (24,000 and 100,000 interferograms) with a self-built radial  
 127 excess-fraction interferometer; and (iii) benchmarking the optical reconstruc-  
 128 tion against 104 tactile reference measurements. Areal parameters are not re-  
 129 ported because the single-image radial extraction does not produce a calibrated  
 130 areal height map. The aim is to assess whether  $L^p$ , bounded-variation, and  
 131 Sobolev seminorms, evaluated alongside ISO 21920-2 profile parameters on the  
 132 same reconstructed residual profile, offer complementary descriptive sensitivity  
 133 to controlled profile perturbations, and to evaluate the within-run repeatability  
 134 of all quantities across five recorded sweeps. No claim of novelty is attached  
 135 to the norms themselves — they are standard tools of functional analysis, and  
 136 the functional-analytic vocabulary is used only as an organising taxonomy; the  
 137 controlled comparison framework is the contribution.

## 138 2. Experimental Details

### 139 2.1. Optical Setup

140 The optical layout of the measurement system is shown in Fig. 1. The  
 141 system is a two-channel modified Michelson interferometer with channels shifted  
 142 by approximately  $90^\circ$  in phase. When an array detector is used instead of a  
 143 point detector, the second channel is not strictly required [39]. In this system it  
 144 was retained because it is useful for object centring and can serve as a control  
 145 channel.

146 The optical setup and the full apparatus-level description were reported in a  
 147 previous study on radial image processing for phase extraction in rough-surface  
 148 interferometry [25]. They are repeated here in condensed form to make the  
 149 present functional-descriptor analysis self-contained and to document the ac-  
 150 quisition conditions behind the FN 111 and FN 101 interferogram archives.

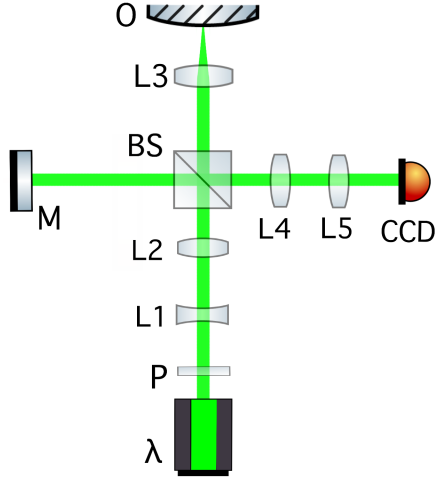


Figure 1: Optical layout of the two-channel modified Michelson interferometer:  $\lambda$  – diode laser (532 nm);  $P$  – linear polariser;  $L$  – lenses;  $L_3$  – microscope objective;  $M$  – reference mirror ( $\lambda/10$  flatness);  $BS$  – non-polarising beamsplitter;  $O$  – object on rotation stage;  $CCD$  – camera.

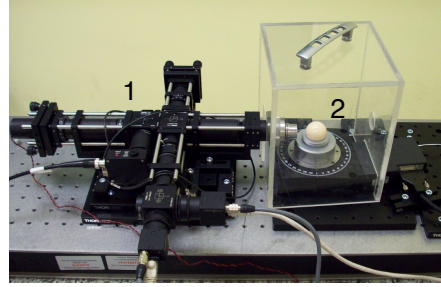


Figure 2: The complete measurement system: (1) modified Michelson interferometer in cage mounting, (2) high-precision rotation stage with the artefact.

151 The system was first tested with the JENOPTIK roundness standard FN 111  
 152 (photographs: Supplementary Fig. S1) ( $Al_2O_3$ ,  $\varnothing = 29.9588$  mm) [40]. A single-  
 153 mode 4.5 mW diode laser with wavelength 532 nm was used as the light source.  
 154 A current-stabilised supply powered the laser to obtain a stable wavelength [41].  
 155 The laser beam passes through a polariser ( $P$  in Fig. 1) to set the polarisation  
 156 of the light at  $45^\circ$  to the setup plane, minimising the difference in reflectivity  
 157 between the two channels.

158 Key components: achromatic lenses  $L_{1,2,4,5}$ , non-polarising beamsplitter

159  $BS$ ,  $\lambda/8$  liquid phase retarder,  $\lambda/10$  reference mirror  $M$ , DIN-4 microscope  
160 objective  $L_3$  (NA 0.10), polarising beamsplitter  $PBS$ , and two Gigabit Ether-  
161 net CCD cameras ( $640 \times 480$  px, 120 fps).

162 The interferometer was built in a cage system for thermal, acoustic and me-  
163 chanical stability. A plexiglass box shields the measurement object, and the  
164 structure is mounted on an area translation stage. A high-precision rotation  
165 stage (encoder accuracy 2.0 mdeg, typical wobble  $\pm 12 \mu\text{rad}$ , typical eccentric-  
166 ity  $\pm 0.40 \mu\text{m}$ ; full specifications: Supplementary Table S1) rotates the object.  
167 Both interferometer and stage are mounted on a honeycomb breadboard with  
168 vibration isolation.

169 Measurements were performed at night (23:00–05:00) to minimise building-  
170 activity disturbances. The setup was thermally and acoustically shielded (plex-  
171 iglass box, wool blanket) and placed on a granite plate on layered wooden-  
172 polystyrene vibration isolation (photographs: Supplementary Fig. S2). Five  
173 semiconductor temperature sensors ( $\pm 0.2^\circ\text{C}$ ) monitored stability; the tempera-  
174 ture near the reference and object arms remained stable to  $\pm 0.2^\circ\text{C}$  over six-hour  
175 acquisitions.

## 176 2.2. Control System and Acquisition Workflow

177 Cameras were controlled by FLIR SDK-based C programs and Bash scripts [42].  
178 Capture time was recorded per frame for angular-position verification. The ro-  
179 tation stage was controlled via USB ASCII commands [43]; a 25 s settling delay  
180 ensured constant velocity before acquisition began. Object centring was per-  
181 formed manually using the two-camera phase-shifted configuration at  $45^\circ$  rota-  
182 tion increments. Fringe evaluation used modified open-source R code [39, 44].  
183 Table 1 lists the acquisition parameters for both archives; five  $400^\circ$  sweeps were  
184 recorded per artefact, with the  $40^\circ$  overlap retained for profile-end comparison.

185 The object was measured in one continuous five-sweep rotation without repo-  
186 sitioning. Each recorded sweep spans  $400^\circ$ ; successive useful  $360^\circ$  segments are  
187 shifted by  $40^\circ$ , giving a natural between-sweep angular offset. The  $40^\circ$  overlap  
188 region provides a model-free repeatability check: RMS height difference between



Table 1: Measurement and sampling parameters of the interferometric acquisition workflow.

| Parameter                                        | FN 111           | FN 101            |
|--------------------------------------------------|------------------|-------------------|
| Rotation speed, $\omega$                         | 0.50 °/s         | 0.08 °/s          |
| CCD frame rate                                   | 6 fps            | 4 fps             |
| Recorded sweeps, $n$                             | 5                | 5                 |
| Total frames, $N$                                | 24,000           | 100,000           |
| Angular step, $\Delta\alpha = \omega/\text{fps}$ | 0.083°/frame     | 0.020°/frame      |
| Frames per degree                                | 12               | 50                |
| Frames per useful 360°                           | 4320             | 18,000            |
| Frames per recorded 400° sweep                   | 4800             | 20,000            |
| Overlap per sweep                                | 40° / 480 frames | 40° / 2000 frames |
| Total acquisition time                           | 4000 s           | 25,000 s          |

consecutive sweeps is 12 nm (FN 111) and 0.18  $\mu\text{m}$  (FN 101), consistent with Table 4.

The frame rates in Table 1 (6 and 4 fps) are far below the 120 fps capability of the cameras and merit comment. The angular step is set by the ratio  $\Delta\alpha = \omega/\text{fps}$ , so angular resolution is governed jointly by rotation speed and frame rate; measurement time is governed by rotation speed alone. The rotation speeds (0.50 and 0.08 °/s) were deliberately conservative: slow, constant-velocity rotation minimises dynamic excitation of the stage and keeps the fringe pattern quasi-static within each exposure, and the settling-time protocol (25 s before acquisition) assumes steady-state velocity. At these speeds, 6 and 4 fps already yield 12 and 50 frames per degree — an oversampling factor of roughly 300 and 1200 relative to the 15 UPR metrological bandwidth — so the frame rate was not the accuracy-limiting factor; increasing it at fixed rotation speed would only add redundant frames. Shorter total acquisition (4,000 and 25,000 s here) would instead require faster rotation with a proportionally higher frame rate, which is technically feasible with the present cameras up to 120 fps. Whether fringe contrast and stage stability are preserved at, say, a tenfold higher speed

has not been tested; the trade-off between acquisition speed, vibration robustness, and per-frame phase-extraction quality is identified as a topic for future work, since measurement time is a critical parameter for industrial implementation.

### 3. Image Processing and Functional-Descriptor Method

#### 3.1. Processing Pipeline

The pipeline follows the radial excess-fraction method of a previous study [25] with added sinusoidal refinement. Each interferogram  $I_i(x, y)$  is decoded independently to a wrapped excess fraction  $\epsilon_i$ ; the sequence is unwrapped chronologically, converted to nanometres ( $z_i = \nu_i \lambda / 2$ ), and reduced by the object-specific LSCI reference model. Sweep boundaries are retained for repeatability analysis.

#### 3.2. Step-by-Step Radial Phase Decoding

For each interferogram  $I_i(x, y)$ , a radial intensity profile about the fringe centre is extracted and B-spline-smoothed; bright-ring maxima are detected in  $D^2$  space, where circular-fringe theory gives  $D_{ij}^2 = c_i(j + \epsilon_i - 1)$  for the  $j$ th ring of diameter  $D_{ij}$ . A linear fit  $D_{ij}^2 = \alpha_i j + \beta_i$  yields the wrapped excess fraction  $\epsilon_i = (1 + \beta_i / \alpha_i) \bmod 1$  (2). Where the radial fringe signal permits ( $R^2 \geq 0.65$ , phase within 0.20 fringe of the ring-diameter estimate), a sinusoidal refinement is applied; otherwise the ring-diameter estimate is retained. The ordered series  $\epsilon_i$  is unwrapped chronologically to a continuous fringe-order coordinate  $\nu_i$ , and the nanometre profile follows from  $z_i = \nu_i \lambda / 2$  with  $\lambda = 532$  nm (one fringe order  $\equiv 266$  nm) (3).

$$D_{ij}^2 = c_i(j + \epsilon_i - 1), \quad (1)$$

$$\epsilon_i = \left(1 + \frac{\beta_i}{\alpha_i}\right) \bmod 1. \quad (2)$$

$$z_i = \nu_i \frac{\lambda}{2}, \quad \lambda = 532 \text{ nm}. \quad (3)$$

### 230 3.3. Topography Profile, Reference Removal and Eccentricity Reduction

231 After phase-to-height conversion, the roundness profile is extracted following  
 232 ISO 12181-1 [45] (which supersedes the withdrawn ISO 4291 [46]). For each 360°  
 233 segment, a least-squares reference circle (LSCI) is fitted to the angular height  
 234 profile  $z_i = h(\theta_i)$  in Cartesian coordinates:

$$(x_i, y_i) = ((R_0 + z_i) \cos \theta_i, (R_0 + z_i) \sin \theta_i), \quad (4)$$

235 where  $R_0 = 14.9794$  mm is the nominal radius of the JENOPTIK FN 111.  
 236 The LSCI centre  $(x_c, y_c)$  minimises  $\sum_i [\sqrt{(x_i - x_c)^2 + (y_i - y_c)^2} - R_0]^2$ , yield-  
 237 ing the centred radial residuals  $\rho_i$ . The LSCI fit is numerically equivalent to  
 238 removing the first harmonic of the angular profile, but it provides the formal  
 239 ISO 4291 reference circle required for roundness metrology. After LSCI cen-  
 240 tring, a phase-correct Gaussian low-pass profile filter with 50% transmission at  
 241 15 undulations per revolution (UPR) is applied to the residual profile, consis-  
 242 tent with the filter specification used for the tactile reference measurements and  
 243 with the Gaussian profile filter defined in ISO 16610-21:2025 [47]. All classical  
 244 and gradient-based descriptors are computed from this LSCI-centred, Gaussian-  
 245 filtered residual  $\rho_i^{(\text{filt})}$ .

246 For the JENOPTIK magnification standard FN 101, the same processing  
 247 chain is applied: a linear trend is removed from the unwrapped phase se-  
 248 quence before height conversion (Sec. 5.1), the LSCI reference circle is fitted  
 249 to the angular profile, and the Gaussian low-pass filter (50% at 15 UPR) is  
 250 applied. Although both artefacts are processed through identical filtering steps,  
 251 the 15 UPR cutoff corresponds to different spatial wavelengths on the circum-  
 252 ference:  $2\pi \times 14.9794 \text{ mm} / 15 = 6.27 \text{ mm}$  for the FN 111 ( $\varnothing = 29.9588 \text{ mm}$ ) and  
 253  $2\pi \times 15 \text{ mm} / 15 = 6.28 \text{ mm}$  for the FN 101 ( $\varnothing \approx 30 \text{ mm}$ ) — effectively identical.  
 254 The angular sampling differs substantially ( $0.083^\circ$  vs.  $0.020^\circ$ , Table 1), but the  
 255 Gaussian filter sigma is set by the UPR cutoff, not by the sampling interval, so  
 256 the filtered profiles are directly comparable.

### 257 3.4. Classical Profile and Roundness Metrics

258 All classical quantities are evaluated on nanometre residuals, not on raw  
 259 intensities, following the profile method of ISO 21920-2 [48], which supersedes  
 260 the earlier ISO 4287 [49]; the amplitude-parameter definitions used here are  
 261 common to both documents, and ISO 4287 is retained as a secondary reference  
 262 because the tactile comparison data and the previous study [25] predate the  
 263 transition. Areal parameters (ISO 25178 [50]) are not reported because the  
 264 single-image radial extraction does not produce a calibrated areal height map.  
 265 For a residual profile or vector of residual pixels  $p$ , the reported profile-style  
 266 roughness descriptors are

$$R_a = \frac{1}{M} \sum_{j=1}^M |p_j - \bar{p}|, \quad R_q = \left[ \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^2 \right]^{1/2}, \quad (5)$$

267 and

$$R_z = \max_j (p_j - \bar{p}) - \min_j (p_j - \bar{p}). \quad (6)$$

268 As implemented, this quantity is the global peak-to-valley height of the full eval-  
 269 uation profile, corresponding to the total-height parameter ( $R_t$  in ISO 21920-2  
 270 terms) rather than the section-averaged  $R_z$ ; the symbol  $R_z$  is retained for conti-  
 271 nuity with the previous study [25]. This implementation detail is relevant to the  
 272 interpretation of Table 4 (Sec. 4.3). The profile skewness  $R_{sk}$  and kurtosis  $R_{ku}$   
 273 — ISO 21920-2 shape parameters that characterise asymmetry and peakedness  
 274 of the height distribution — are also reported:

$$R_{sk} = \frac{1}{R_q^3} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^3, \quad R_{ku} = \frac{1}{R_q^4} \frac{1}{M} \sum_{j=1}^M (p_j - \bar{p})^4. \quad (7)$$

275  $R_{sk} = 0$  for a symmetric distribution;  $R_{sk} < 0$  indicates a predominance of val-  
 276 leys over peaks.  $R_{ku} = 3$  for a normal distribution;  $R_{ku} > 3$  indicates a spikier  
 277 (leptokurtic) profile. These are the only ISO 21920-2 amplitude-distribution pa-  
 278 rameters that capture distribution shape beyond amplitude averages, and their  
 279 inclusion allows a direct test of whether gradient-based descriptors provide in-  
 280 formation beyond even the best ISO shape parameters. For the eccentricity-

281 corrected ceramic angular profile  $q_C(\theta)$ ,

$$RONt = \max_{\theta} q_C(\theta) - \min_{\theta} q_C(\theta), \quad RONq = \left[ \frac{1}{2\pi} \int_0^{2\pi} q_C^2(\theta) d\theta \right]^{1/2}. \quad (8)$$

282 These are profile/form quantities in nanometres. B-spline pre-smoothing (20  
283 degrees of freedom,  $\approx 18^\circ$  period) stabilises the LSCI fit; the subsequent Gaus-  
284 sian low-pass filter (50% at 15 UPR, ISO 16610-21:2025) defines the metrolog-  
285 ical bandwidth. All descriptors are computed from the same Gaussian-filtered  
286 residual.

### 287 3.5. Functional Gradient-Based Descriptors

288 The concept of a complete normed vector space originates from the founda-  
289 tional work of Banach [51]. For discrete profile data of finite length  $N$ , the vector  
290 space  $\mathbb{R}^N$  equipped with any norm is a Banach space; the present work does  
291 not invoke infinite-dimensional completeness but draws on the taxonomy of  $L^p$ ,  
292 Sobolev, and bounded-variation norms that Banach’s framework systematised.  
293 For a centred residual profile  $p(s)$  and  $1 \leq p < \infty$ ,

$$\|p\|_p = \left( \frac{1}{L} \int_0^L |p(s)|^p ds \right)^{1/p}, \quad \|p\|_\infty = \max_{s \in [0, L]} |p(s)|. \quad (9)$$

294 The reported functional descriptors include  $L^1$ ,  $L^2$ ,  $L^\infty$ , the one-dimensional  
295 Sobolev  $W^{1,2}$  seminorm and the bounded-variation (BV) total variation,

$$|p|_{W^{1,2}} = \left( \int_0^L \left| \frac{dp}{ds} \right|^2 ds \right)^{1/2}, \quad TV(p) = \int_0^L \left| \frac{dp}{ds} \right| ds. \quad (10)$$

296 In discrete form, for a uniformly sampled profile  $p_j$  ( $j = 1, \dots, N$ ) with angular  
297 spacing  $\Delta\alpha$ ,

$$|p|_{W^{1,2}} \approx \left( \Delta\alpha \sum_{j=1}^{N-1} \left( \frac{p_{j+1} - p_j}{\Delta\alpha} \right)^2 \right)^{1/2}, \quad TV(p) \approx \sum_{j=1}^{N-1} |p_{j+1} - p_j|. \quad (11)$$

298 Both  $W^{1,2}$  and BV density carry sampling-interval dependence; “nm sample<sup>-1</sup>”  
299 is retained in the result tables for the fixed-acquisition setup, and a standardised  
300 angular-unit convention for cross-instrument comparison is defined in Sec. 3.6.

301 The curvature RMS (root-mean-square of the discrete second derivative;  
 302 reported in Supplementary Table S2) carries a  $(\Delta\alpha)^{-2}$  sampling dependence and  
 303 should not be compared across instruments with different angular resolution.

### 304 3.6. Angular-Unit Convention for Sampling-Dependent Descriptors

305 Gradient-based descriptors evaluated on discretely sampled profiles depend  
 306 on the sampling interval: the discrete total variation of an unfiltered profile  
 307 grows as the sampling grid refines (noise contributes gradient content at every  
 308 scale), so raw per-sample values are instrument-specific. For meaningful cross-  
 309 instrument comparison the following convention is proposed and used here.

- 310 1. **Band-limit first.** The profile is filtered with a declared phase-correct  
 311 Gaussian profile filter (ISO 16610-21:2025) at a stated cutoff in undu-  
 312 lations per revolution (here 50% transmission at 15 UPR). After band-  
 313 limiting, the profile derivative is bounded and the discrete sums converge  
 314 to the continuous integrals as  $\Delta\alpha \rightarrow 0$ ; the descriptor then characterises  
 315 the declared bandwidth rather than the sampling grid.
- 316 2. **Sample adequately.** The angular sampling must resolve the declared  
 317 bandwidth with margin; at least ten samples per cutoff undulation are  
 318 recommended. Here the FN 111 and FN 101 profiles carry  $\approx 290$  and  
 319  $\approx 1200$  samples per 15-UPR undulation, so discretisation error is negligible.
- 320 3. **Report in angular units.** The BV density is reported as total varia-  
 321 tion per unit angle,  $BV_\theta = TV(p)/\Theta$  in  $\text{nm deg}^{-1}$  ( $\Theta = 360^\circ$ ), and the  
 322 gradient content as the RMS angular gradient  $g_{\text{rms}} = [\frac{1}{N-1} \sum_j (p_{j+1} -$   
 323  $p_j)/\Delta\alpha)^2]^{1/2}$  in  $\text{nm deg}^{-1}$ , together with the filter cutoff. For compari-  
 324 son across artefacts or instruments of *different diameter*, the angular unit  
 325 is additionally converted to an arc-length unit by dividing by the arc  
 326 length per degree,  $\pi D/360$ ; the BV density is then  $BV_s = TV(p)/(\pi D)$  in  
 327  $\text{nm mm}^{-1}$ . The arc-length form is the diameter-independent unit and is  
 328 the one recommended for cross-instrument reporting, since  $\text{nm deg}^{-1}$  still  
 329 scales with the circumference of the specific artefact.

330 A note on physical dimensions clarifies why this is necessary and is itself part  
 331 of the answer to the redundancy question (Sec. 5.3). Amplitude descriptors ( $R_a$ ,  
 332  $R_q$ ,  $L^1$ ,  $L^2$ ,  $L^\infty$ ,  $RONt$ ) have the dimension of length (nm) and are sampling-  
 333 independent. Gradient descriptors are dimensionally distinct: BV density and  
 334 the RMS gradient are height-per-length ( $\text{nm mm}^{-1}$ ), i.e. dimensionless slopes,  
 335 while the  $W^{1,2}$  seminorm as defined here is  $\text{height} \times \text{length}^{-1/2}$ . This dimen-  
 336 sional separation is the reason the gradient layer carries information that no  
 337 amplitude parameter can encode, and also the reason its raw per-sample value  
 338 must be qualified by a sampling and filter declaration.

339 For the present data, the per-sample values in Table 4 convert as follows.  
 340 FN 111 ( $\Delta\alpha = 0.083^\circ$ ): BV density  $0.263 \text{ nm sample}^{-1} \rightarrow 3.2 \text{ nm deg}^{-1} \rightarrow$   
 341  $12.1 \text{ nm mm}^{-1}$ ; RMS gradient  $0.310 \text{ nm sample}^{-1} \rightarrow 3.7 \text{ nm deg}^{-1} \rightarrow 14.2 \text{ nm mm}^{-1}$ .  
 342 FN 101 ( $\Delta\alpha = 0.020^\circ$ ): BV density  $6.58 \text{ nm sample}^{-1} \rightarrow 329 \text{ nm deg}^{-1} \rightarrow$   
 343  $1.26 \mu\text{m mm}^{-1}$ ; RMS gradient  $8.76 \text{ nm sample}^{-1} \rightarrow 438 \text{ nm deg}^{-1} \rightarrow 1.67 \mu\text{m mm}^{-1}$ ,  
 344 all at the 15-UPR cutoff. The two artefacts have almost identical circumference  
 345 ( $\pi D \approx 94.1$  and  $94.2 \text{ mm}$ ), so the large difference in arc-length gradient con-  
 346 tent ( $12$  versus  $1257 \text{ nm mm}^{-1}$ ) reflects a genuine difference in surface variation  
 347 rather than a unit artefact. Because each conversion is a fixed multiplicative  
 348 factor for a given acquisition, relative uncertainties and relative defect responses  
 349 are identical in every unit system. Adoption of gradient descriptors in reference  
 350 software for roundness evaluation would require exactly this pair of declarations  
 351 — filter cutoff and arc-length units — which parallels the established practice  
 352 of declaring the filter for  $RONt$ .

353 An empirical demonstration that this convention removes the sampling de-  
 354 pendence is obtained by decimating the FN 111 residual of session A from 12  
 355 to 1 samples per degree (a factor of 12) and recomputing the descriptors with  
 356 the 15-UPR Gaussian filter matched to each sampling. The band-limited BV  
 357 density and RMS gradient change by less than 0.5% across the full decimation  
 358 range, whereas the unfiltered total variation drops by a factor of 3.3 over the  
 359 same range. This confirms that the declared filter — not the sampling grid  
 360 — defines the descriptor value, and shows that the recommendation of at least

ten samples per cutoff undulation is conservative: even 24 samples per cutoff undulation reproduce the band-limited values to within 0.5%.

### 3.7. Uncertainty Evaluation

Each dataset contains five  $400^\circ$  sweeps recorded in a single continuous rotation without stopping or repositioning. The five useful  $360^\circ$  segments are extracted from this continuous sequence; because each recorded sweep spans  $400^\circ$ , successive useful revolutions are shifted by  $40^\circ$  on the object. These segments are not strictly independent — they share the same continuous-run conditions (laser warm-up state, stage-bearing temperature, ambient drift). The standard deviation across sweeps therefore reflects within-run repeatability rather than between-run reproducibility. For a descriptor  $X$  evaluated on each sweep, the mean value is  $\bar{X}$  and the within-run standard uncertainty is estimated as

$$u_{\text{within}}(X) = \frac{s_X}{\sqrt{n}}, \quad n = 5, \quad (12)$$

where  $s_X$  is the sample standard deviation across the five sweeps. The reported expanded uncertainty is

$$U_{95}(X) = t_{0.975, n-1} u_{\text{within}}(X). \quad (13)$$

With  $n = 5$ ,  $t_{0.975, 4} = 2.776$ ; the  $\chi^2_4$ -based 95% CI for the standard deviation spans  $[0.6s_X, 2.9s_X]$ , so  $U_{95}$  is indicative rather than definitive. Because the five segments share continuous-run conditions (laser state, stage temperature, ambient drift), the  $1/\sqrt{n}$  factor in  $u_{\text{within}}$  may underestimate the true uncertainty of the mean; all reported uncertainties are *lower bounds*. An indicative type-B budget for the principal systematic terms is given in Sec. 3.8; a complete GUM-compliant budget [52] with experimentally characterised input distributions, and a between-run reproducibility study with independent setups, thermal cycles, and physical re-centring of the artefact, remain future work (Sec. 5.6). With  $n = 5$  sweeps and the observed within-run variability, the minimum detectable relative difference between two descriptors (80% power,  $\alpha = 0.05$ , two-sided paired  $t$ -test) is approximately 30% for the FN 111 and 5% for the FN 101, consistent with the exploratory character of the descriptor comparison. A per-sweep



388 trend of FN 111  $RONt$  values gives a slope of  $8.4 \pm 4.1$  nm sweep $^{-1}$  ( $p = 0.12$ ),  
 389 consistent with zero drift, though the sweep-to-sweep scatter cautions that  $U_{95}$   
 390 may underestimate long-term reproducibility.

### 391 3.8. Indicative Type-B Uncertainty Budget

392 Although a complete GUM budget requires between-run data that the present  
 393 single-rotation acquisition cannot supply, the principal type-B contributions can  
 394 be bounded from component specifications and pipeline properties. Table 2  
 395 summarises the estimates for the FN 111  $RONt$  evaluation; the same relative  
 396 arguments apply to the FN 101.

397 *Laser wavelength stability.* The height scale  $z = \nu\lambda/2$  inherits the relative  
 398 wavelength error: a  $\pm 1$  nm tolerance on the 532 nm diode module (rectangu-  
 399 lar distribution) gives a relative standard uncertainty of 0.11%, i.e. 0.3 nm on  
 400  $RONt = 278$  nm (and 46 nm on the FN 101  $RONt = 42.07$   $\mu$ m, within its  
 401  $\pm 0.48$   $\mu$ m repeatability interval). Wavelength error acts as a pure scale factor  
 402 and cannot create or mask localised features.

403 *Refractive index of air.* The in-air wavelength varies with temperature and  
 404 pressure by  $\sim 10^{-6}$  per  $1^\circ\text{C}$  or 3 hPa; with the observed  $\pm 0.2^\circ\text{C}$  stability the  
 405 contribution is below  $10^{-5}$  relative, i.e.  $< 0.01$  nm. Negligible.

406 *Abbe error and stage wobble.* The specified axis wobble ( $\pm 12$   $\mu$ rad) enters  
 407 through two mechanisms: a cosine-type radial error  $R\theta_w^2/2 \approx 1$  pm, and a lateral  
 408 beam walk  $R\theta_w \approx 0.18$   $\mu$ m on the circumference, equivalent to an angular  
 409 mislocation of  $0.0007^\circ$ , which maps through the local band-limited gradient  
 410 ( $\approx 3.2$  nm deg $^{-1}$ , Sec. 3.6) to  $\approx 0.002$  nm. Both are negligible after LSCI  
 411 first-harmonic removal.

412 *Camera nonlinearity and EFM localisation.* Intensity nonlinearity perturbs  
 413 the radial fringe-peak positions; its effect is bounded empirically by the per-  
 414 frame excess-fraction uncertainty of  $\pm 0.05$  fringe (13 nm). The 15-UPR band  
 415 limit averages  $\approx 290$  frames per resolved undulation, reducing the per-point  
 416 contribution to  $\approx 0.8$  nm and the peak-to-valley contribution to  $\approx 1.1$  nm.

417 *Long-term drift and phase noise floor (measured).* An independent static

418 stability session, recorded with the same interferometer, camera settings and  
 419 frame count as the FN 111 acquisition (24,000 frames over 4,000 s) but with the  
 420 object not rotating, provides a direct experimental bound on slow instrument  
 421 drift under realistic laboratory conditions. The decoded excess-fraction sequence  
 422 spans only 98 nm over the full session, with an end-to-end linear drift of  $-82$  nm  
 423 ( $-1.1$  nm min $^{-1}$ ), i.e.  $\approx 15$  nm over one 800 s sweep-equivalent interval; after  
 424 linear detrending the residual has 9.4 nm standard deviation (45 nm peak-  
 425 to-valley), and the frame-to-frame phase noise floor is 2.1 nm (0.008 fringe).  
 426 Two conclusions follow. First, the measured static noise floor is well below  
 427 the conservative  $\pm 0.05$ -fringe EFM bound used above, confirming that bound  
 428 as conservative. Second, the slow drift maps predominantly to low harmonic  
 429 orders of the angular profile, where it is largely absorbed by the LSCI first-  
 430 harmonic removal, and its sweep-to-sweep manifestation is already contained  
 431 in the type-A scatter; the drift bound is therefore reported as an informative  
 432 constraint rather than added in quadrature.

433 *Form removal.* The LSCI reference (first-harmonic removal) is exact only  
 434 in the limaçon approximation; the second-order term  $a^2/(2R)$  for residual cen-  
 435 tring eccentricity  $a$  leaks into the second harmonic of the residual. Manual  
 436 centring leaves  $a$  of the order of tens of micrometres (the shared first-harmonic  
 437 model of Sec. 5.2.1, with its 63 nm/0.18% per-sweep amplitude variation, im-  
 438 plies  $a \approx 35$   $\mu$ m); for  $a = 15$ – $35$   $\mu$ m the term evaluates to 7–41 nm. Because  
 439 this contribution is common to all five sweeps, it does not appear in the type-A  
 440 scatter and must be carried as a bounded systematic term. Its magnitude is  
 441 consistent with the 34 nm difference between the optical  $RONt$  ( $278 \pm 22$  nm)  
 442 and the certified tactile value ( $244 \pm 52$  nm), a difference that the  $E_n = 0.61$   
 443 comparison shows to be within the combined uncertainties.

444 The quadrature combination is dominated by the type-A term, so the re-  
 445 ported  $U_{95}$  values are essentially unchanged by the enumerated random type-B  
 446 terms; the budget nevertheless shows that the leading risks are the form-removal  
 447 systematic and the between-run effects bounded in Sec. 3.9, not the enumerated  
 448 instrument terms.

Table 2: Indicative type-B uncertainty budget for the FN 111  $RONt$  evaluation. Random contributions are standard uncertainties combined in quadrature with the type-A term; the form-removal term is a bounded systematic (bias) contribution common to all sweeps and is reported separately.

| Source                                              | Basis                                               | Contribution to $RONt$         |
|-----------------------------------------------------|-----------------------------------------------------|--------------------------------|
| Within-run repeatability (type A)                   | $s/\sqrt{5}$ , five sweeps                          | 7.9 nm                         |
| Laser wavelength ( $\pm 1$ nm, rect.)               | module specification                                | 0.3 nm                         |
| Air refractive index ( $\pm 0.2^\circ\text{C}$ )    | Edlén-type sensitivity                              | $< 0.01$ nm                    |
| Abbe error / axis wobble ( $\pm 12 \mu\text{rad}$ ) | stage specification, geometry                       | $< 0.01$ nm                    |
| Camera nonlinearity / EFM localisation              | $\pm 0.05$ fringe per frame, band-limited averaging | 1.1 nm                         |
| Long-term drift (measured, static session)          | $-1.1 \text{ nm min}^{-1}$ ; $\leq 15$ nm per sweep | informative bound <sup>†</sup> |
| Combined standard uncertainty $u_c$ (random)        | quadrature                                          | 8.0 nm                         |
| Expanded $U_{95}$ (random, $t_{0.975,4}$ )          |                                                     | 22 nm                          |
| Form removal (limaçon second order)                 | $a^2/(2R)$ , $a = 15\text{--}35 \mu\text{m}$        | $\leq 41$ nm (syst. bound)     |

<sup>†</sup>Measured in an independent static session (Sec. 3.8); manifested in the type-A scatter and largely removed with the first harmonic, hence not added in quadrature.

### 3.9. Between-Run Reproducibility of Independent Sessions

Two further independent acquisition sessions of the FN 111 were processed through the identical pipeline and descriptor computation (per-sweep LSCI-centred residual, circular Gaussian low-pass filter at 15 UPR) to assess between-run reproducibility with the archived material. The primary archive reported in Tables 1 and 4 is denoted session A. Session B (0.50°/s, 6 fps, 24,000 frames) repeats its acquisition parameters; session C (0.64°/s, 8 fps, 25,000 frames, 0.08°/frame) uses a different rotation speed and frame rate at nearly the same angular sampling. Sessions B and C were recorded on different days with the artefact re-mounted and re-centred. Table 3 reports the relative between-session differences and the  $E_n$  compatibility statistic,  $E_n = |\Delta|/(2\sqrt{u_A^2 + u_B^2})$ , with the standard uncertainties of the per-sweep means ( $n = 5$ ).

Table 3: Between-session comparison for the FN 111. Session A is the primary archive reported in Tables 1–3; sessions B and C are independent repeat acquisitions (B: same parameters; C: 0.64°/s, 8 fps) with the artefact re-mounted and re-centred. Relative differences and  $E_n$  compatibility values are reported;  $E_n < 1$  indicates agreement within the combined standard uncertainties of the per-sweep means.

| Descriptor      | B vs. A    |       | C vs. A    |       |
|-----------------|------------|-------|------------|-------|
|                 | rel. diff. | $E_n$ | rel. diff. | $E_n$ |
| $R_a$           | −2.4%      | 0.35  | +7.0%      | 1.45  |
| $R_q$           | −2.0%      | 0.23  | +8.8%      | 1.51  |
| $RONt$          | −2.9%      | 0.23  | +12.3%     | 1.16  |
| $L^\infty$      | −0.1%      | 0.00  | +11.6%     | 0.79  |
| BV density      | −4.0%      | 0.99  | +4.4%      | 1.25  |
| $W^{1,2}$ grad. | −3.8%      | 0.88  | +4.6%      | 1.40  |

For session B, every descriptor agrees with session A within its combined uncertainty ( $|E_n| < 1$ ): the amplitude parameters differ by less than 3% and the gradient descriptors by about 4% (BV density,  $E_n = 0.99$ , is the largest), showing that BV density and the  $W^{1,2}$  seminorm are between-run stable to the same order as the classical parameters under identical acquisition parameters. For session C, the amplitude descriptors differ by 7–12% ( $E_n$  up to 1.5); the

467 *RONt* difference of 42 nm is comparable to the  $\leq 41$  nm form-removal bound  
468 of Sec. 3.8, indicating that the dominant between-session variation arises from  
469 re-centring (first-harmonic leakage) rather than from the change of rotation  
470 speed or frame rate. The gradient descriptors change by +4% (per-sample) to  
471 +9% (angular units), tracking the shared amplitude change with no sampling-  
472 induced anomaly. Together with the static-session drift bound (Sec. 3.8), these  
473 results establish the first quantitative between-run stability statement for the  
474 gradient-based descriptors; a full multi-session campaign remains future work  
475 (Sec. 5.6).

### 476 3.10. Computational Implementation

477 The reconstruction workflow (`scripts/reconstruct_radial_profile_sequence.`  
478 `py`) decodes  $\epsilon_i$  from each interferogram, unwraps the sequence, converts to  
479 nanometres, applies LSCI fitting and Gaussian low-pass filtering (50% at 15 UPR,  
480 ISO 16610-21:2025), and exports CSV/JSON metrics with sweep-repeatability  
481 uncertainty. Vector PDF figures document each processing stage. The pipeline  
482 is invoked as `PYTHONPATH=src.venv/bin/pythonscripts/reconstruct_radial_`  
483 `profile_sequence.py` with dataset-specific parameters (full command in the  
484 data repository). Per-image excess-fraction decoding and unwrapping require  
485 approximately 0.5 ms per frame (measured on an Intel Core i7-9700K, single-  
486 threaded Python 3.9), so the full 124,000-frame archive processes in under two  
487 minutes on consumer hardware.

## 488 4. Results

489 The results follow the updated radial sequence route only. Each raw im-  
490 age is first decoded to a wrapped excess fraction from its radial fringe pat-  
491 tern. The ordered excess-fraction series is then unwrapped in chronological  
492 order, converted to nanometres, and corrected for eccentricity. Finally, classi-  
493 cal profile/roundness descriptors and functional gradient-based descriptors are  
494 computed from the same nanometre residual profiles.

#### 4.1. Radial Phase Decoding and Nanometre Profile Reconstruction

The FN 111 folder contained 24,000 interferograms and the steel folder contained 100,000 interferograms. The reported run used full-resolution images, the central radial aperture of 240 px, B-spline-smoothed peak detection, continuity-aware EFM peak-subset tracking and an EFM prominence threshold of 0.04 of the radial fringe-signal range. All five recorded sweeps were processed.

Figs. S3–S6 (single-image EFM and sequence processing steps) are provided in the Supplementary Materials.

#### 4.2. Roughness, Roundness and Functional Descriptors

Table 4 gives the numerical comparison after LSCI reference circle and Gaussian low-pass filtering (50% at 15 UPR). Both artefacts are processed through the same filtering chain. Each descriptor is reported as the full-sequence value together with the 95% expanded type-A repeatability uncertainty estimated from the five sweeps. Per-sweep values for all descriptors are provided in Supplementary Table S3.

#### 4.3. Coincidence of $R_z$ with $RONt$ and $R_q$ with $RONq$

The identical values of  $R_z$  and  $RONt$  (and of  $R_q$  and  $RONq$ ) for both artefacts in Table 4 are expected by construction, not coincidental. In this pipeline both descriptor families are evaluated on the same LSCI-centred, Gaussian-filtered closed residual profile (Sec. 3.3; harmonic decomposition of the residual: Supplementary Fig. S19): the roughness-style  $R_z$  is the global peak-to-valley height of that profile (i.e., the total height  $Rt$ ; see the implementation note in the definition), while  $RONt$  is the peak-to-valley roundness deviation of the very same profile, so the two coincide exactly; the RMS pair  $R_q/RONq$  coincides for the same reason. The two families would separate if profile texture were evaluated on a different bandwidth than form (e.g., after  $\lambda c$ -based separation of roughness from waviness in the ISO 21920 sense) or on an independent linear trace. Both are nevertheless reported because they belong to different ISO

Table 4: Full-resolution radial EFM reconstruction from raw interferograms (LSCI + Gaussian low-pass, 50% at 15 UPR, ISO 16610-21:2025). Uncertainties are 95% expanded within-run repeatability estimates from five sweeps ( $n = 5$ ,  $t_{0.975,4} = 2.776$ ). Units: nm for all descriptors except BV density and  $W^{1,2}$  gradient ( $\text{nm sample}^{-1}$ ). Certified *RONt*: FN 111  $244 \pm 52$  nm (Hommel Roundscan 535,  $n = 104$ ); FN 101  $41.94 \mu\text{m}$  (form tester,  $n = 12$ ). The FN 101 certificate does not report an expanded uncertainty; therefore only the percentage difference is reported for this artefact rather than an  $E_n$  score.

| Descriptor                  | FN 111                                     | FN 101                                   |
|-----------------------------|--------------------------------------------|------------------------------------------|
| $R_a$                       | $(66.3 \pm 4.7) \text{ nm}$                | $(8.525 \pm 0.115) \mu\text{m}$          |
| $R_q$                       | $(77.6 \pm 5.0) \text{ nm}$                | $(10.376 \pm 0.090) \mu\text{m}$         |
| $R_z$                       | $(278 \pm 22) \text{ nm}$                  | $(42.07 \pm 0.48) \mu\text{m}$           |
| <i>RONt</i>                 | $(278 \pm 22) \text{ nm}$                  | $(42.07 \pm 0.48) \mu\text{m}$           |
| <i>RONq</i>                 | $(77.6 \pm 5.0) \text{ nm}$                | $(10.376 \pm 0.090) \mu\text{m}$         |
| $L^1$ mean                  | $(66.3 \pm 4.7) \text{ nm}$                | $(8.525 \pm 0.115) \mu\text{m}$          |
| $L^2$ RMS                   | $(77.6 \pm 5.0) \text{ nm}$                | $(10.376 \pm 0.090) \mu\text{m}$         |
| $L^\infty$                  | $(150 \pm 12) \text{ nm}$                  | $(21.97 \pm 0.42) \mu\text{m}$           |
| BV total                    | $(1.137 \pm 0.062) \mu\text{m}$            | $(118.4 \pm 6.2) \mu\text{m}$            |
| BV density                  | $(0.263 \pm 0.014) \text{ nm sample}^{-1}$ | $(6.58 \pm 0.35) \text{ nm sample}^{-1}$ |
| $W^{1,2}$ grad.             | $(0.310 \pm 0.015) \text{ nm sample}^{-1}$ | $(8.76 \pm 0.43) \text{ nm sample}^{-1}$ |
| <i>ISO shape parameters</i> |                                            |                                          |
| $R_{sk}$ (skewness)         | $-0.09 \pm 0.04$                           | $0.00 \pm 0.09$                          |
| $R_{ku}$ (kurtosis)         | $1.98 \pm 0.21$                            | $2.51 \pm 0.08$                          |
| <i>Derived norm ratios</i>  |                                            |                                          |
| $L^4/L^1$ peakiness         | $1.39 \pm 0.08$                            | $1.53 \pm 0.03$                          |
| <i>External validation</i>  |                                            |                                          |
| $E_n$ score                 | 0.61                                       | $< 1$ (0.3% diff.; cf. Fig. S12)         |

frameworks (profile texture versus roundness) with different reference conventions, and because their exact equality serves as a verification that all descriptors share one evaluation profile — the design premise of the like-for-like comparison.

#### 4.4. *Classical-Functional-Descriptor Comparison*

The exact equalities  $R_a = L^1$  and  $R_q = L^2$  in Table 4 are definitional — each pair is the same formula applied to the same centred residual — and are reported purely as a computational cross-check that both descriptor families are evaluated on identical data, not as an empirical finding (see Sec. 5.3). For the FN 111, BV total variation carries 1.2% relative expanded uncertainty versus 5.5% for  $R_a$  and 8.5% for  $R_z$ , suggesting that gradient-based descriptors may exhibit metrological stability comparable to or better than peak-referenced roughness parameters in the tested datasets. Comparative bar charts and uncertainty analyses are provided in the Supplementary Materials (Figs. S7–S8). Ceramic and steel profile reconstruction plots are shown in Supplementary Figs. S9–S10.

The steel roundness polar representation (FN 101, Flick flat) is provided in the Supplementary Materials (Fig. S11).



JENOPTIK FN~111 — Roundness deviation  
 $RONt = 278$  nm (5 sweeps, native angular coordinates)

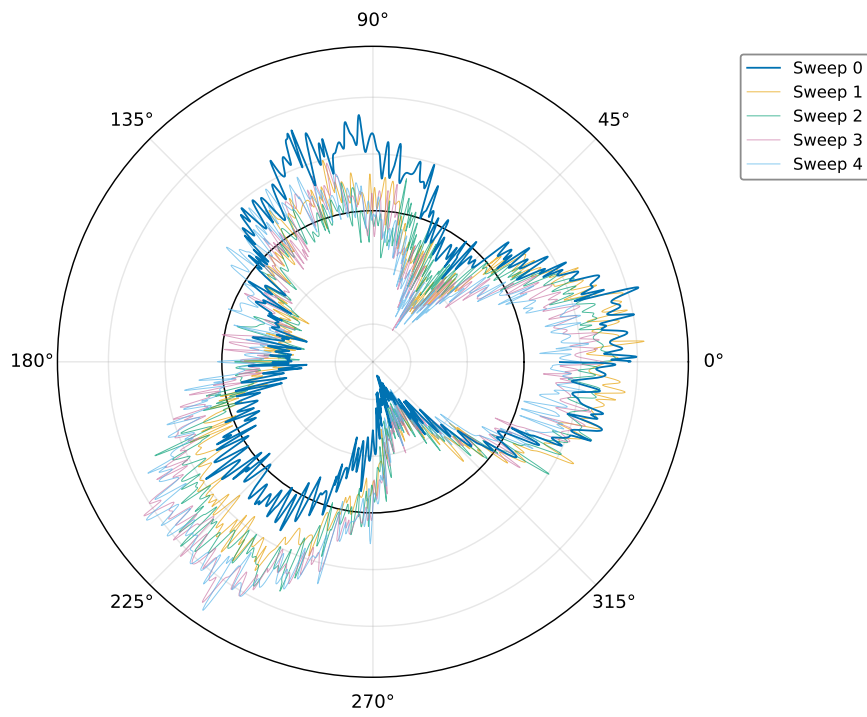


Figure 3: Ceramic roundness in polar coordinates (LSCI + Gaussian low-pass, 50% at 15 UPR). Five sweeps at native positions.  $RONt = 278 \pm 22$  nm.

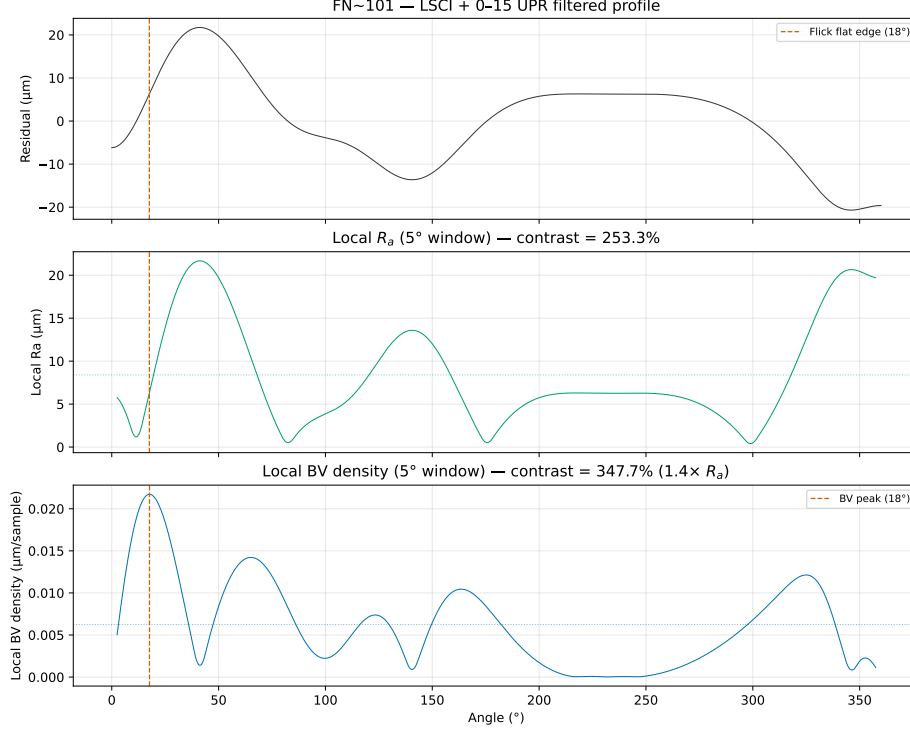


Figure 4: Flick flat detection on the FN 101 magnification standard. BV density (bottom) shows a clear spike at the Flick flat angular position, whereas  $R_a$  computed in a sliding window (top) shows no detectable response at the reconstruction resolution. The sliding-window width was matched to the certified angular extent of the Flick flat as specified in the DAkKS-DKD certificate (Art. 532 528); identical window parameters were used for both descriptors.

539 The Flick flat produces a clear response in BV density while showing no  
 540 detectable response in a sliding-window  $R_a$  analysis at the reconstruction reso-  
 541 lution (Fig. 4), illustrating the potential complementary sensitivity of gradient-  
 542 based descriptors to localised geometric features.

## 543 5. Discussion

### 544 5.1. Confirmed Results

545 The analysis rests on a key metrological distinction: an interferogram is  
 546 an intensity measurement, not a phase map and not a height map. The half-

wavelength relation  $z = \nu\lambda/2$  becomes valid only after each image has been decoded to a wrapped excess fraction and the ordered excess-fraction sequence has been unwrapped to the continuous fringe-order coordinate  $\nu$ . This is why the reported comparison uses only reconstructed nanometre profiles, not raw greyscale images and not areal height maps.

The most sensitive part of the workflow is the EFM localisation of bright-ring maxima, with an excess-fraction uncertainty of approximately  $\pm 0.05$  fringe orders (tightened to  $\pm 0.03$  by sinusoidal refinement in 72% of FN 111 frames). A single  $\epsilon_i$  error of 0.05 fringe corresponds to 13 nm, sub-dominant to the 85 nm residual-profile RMS. The chronological unwrapping criterion rejects phase jumps exceeding 0.35 fringe. The median number of detected rings is three for both artefacts within the 240 px central aperture. The five 400° ceramic sweeps require a geometric correction: successive useful 360° intervals are shifted by 40° because the physical object angle advances continuously; without this correction, the LSCI fit appears inconsistent even when phase extraction within each sweep is repeatable.

The linear drift of  $0.0246 \text{ fringe frame}^{-1}$  in the steel unwrapped phase sequence (versus 0.00021 for the FN 111) is attributed to cumulative bias in the global unwrapping algorithm rather than thermal effects or axial runout [53]. The lower fringe contrast of the FN 101 interferograms — likely due to surface finish — reduces per-frame EFM quality; the pipeline applies automatic linear detrending where needed.

## 5.2. Interpretation

### 5.2.1. Eccentricity removal and wobble verification

Roundness descriptors are computed after LSCI reference-circle fitting (Sec. 3.3), which removes the first harmonic. A shared model across all five sweeps yields  $R^2 = 0.99991$  with per-sweep amplitude variation of 63 nm (0.18%); the manufacturer-specified wobble ( $\pm 12 \mu\text{rad}$ ) bounds the observed variation an order of magnitude below  $RONt = 278 \text{ nm}$ . The polar representation (Fig. 3) shows five

576 overlaid sweeps repeatable to  $\pm 50$  nm, illustrating that the residual is domi-  
 577 nated by low-order form deviation rather than random roughness.

### 578 5.3. Comparison with ISO Parameters

579 A central question of this work is whether gradient-based functional descrip-  
 580 tors provide additional descriptive sensitivity alongside classical ISO parameters  
 581 when applied to the same reconstructed profiles. To answer this rigorously, every  
 582 descriptor was computed on the same reconstructed nanometre residual profiles.

583 **Mathematical equivalences.** The  $L^1$  norm is defined as  $\frac{1}{N} \sum |p_i - \bar{p}|$ ,  
 584 which is identical to the definition of  $R_a$ . The  $L^2$  norm is  $\sqrt{\frac{1}{N} \sum (p_i - \bar{p})^2}$ ,  
 585 identical to  $R_q$ . Computing both on the same centred residual yields *exact*  
 586 *numerical equality*:  $R_a - L^1 = 0$  and  $R_q - L^2 = 0$  to machine precision for  
 587 both artefacts (Table 4). These equalities are not empirical findings; they are  
 588 definitional consequences. Labelling the arithmetic mean of absolute deviations  
 589 as  $L^1$  rather than  $R_a$  provides no new metrological information and should not  
 590 be presented as a methodological advance.

591 It is emphasised that neither the  $L^p$  norms nor the Sobolev and BV semi-  
 592 norms are presented as novel mathematical constructs; they are standard tools  
 593 whose definitions are restated for completeness. Their value in the present con-  
 594 text is threefold. First, the functional-analytic taxonomy places ISO parameters  
 595 at specific positions ( $L^1$ ,  $L^2$ ,  $L^\infty$ ) within a larger space of possible descriptors,  
 596 revealing that descriptors outside the ISO set — BV density,  $W^{1,2}$  seminorm  
 597 — respond differently to the same perturbations. Second, the synthetic-defect  
 598 injection methodology provides a controlled, like-for-like comparison framework  
 599 that can be applied to any descriptor computed from the same residual profile;  
 600 the framework itself, rather than any individual norm, constitutes the method-  
 601 ological contribution. Third, the Flick flat on the certified FN 101 standard pro-  
 602 vides a concrete existence proof: a real, DAKS-DKD-certified artefact whose  
 603 defining geometric feature is detected unambiguously by BV density while re-  
 604 maining below the detection threshold of sliding-window  $R_a$  (Fig. 4).

605 **Extrema:  $R_z$  versus  $L^\infty$ .**  $R_z/L^\infty \approx 2$  for both artefacts, consistent with

606 symmetric positive and negative excursions.  $L^\infty$  captures the single largest  
 607 one-sided excursion; the ratio serves as a simple symmetry indicator.

608 **Variation-based descriptors.** A controlled sensitivity analysis (Supple-  
 609 mentary Table S2) examined five synthetic defect types injected into the FN 111  
 610 residual profile. Amplitude-averaging descriptors ( $R_a$ ,  $R_q$ ,  $R_z$ ) showed no de-  
 611 tectable change at the reconstruction resolution — a single 100 nm spike changed  
 612  $R_a$  by less than 0.05%, below the per-frame phase-extraction precision — while  
 613 BV density increased by +18% and the  $W^{1,2}$  gradient seminorm by +622% for  
 614 the same spike (Fig. 5). ISO shape parameters responded to asymmetry ( $R_{sk}$   
 615 changed by −1104% for a step) but not to symmetric localised features.  $R_{sk}$   
 616 and BV density appear complementary in the tested scenarios:  $R_{sk}$  responds to  
 617 asymmetry, BV density responds to localised gradients.

618 **Translation of synthetic responses to realistic defects.** The injected  
 619 defects are idealised (isolated spike, uniform scratch, pure sinusoid), whereas  
 620 real manufacturing defects have complex, random morphologies. The synthetic  
 621 responses nevertheless translate to other backgrounds through a simple scal-  
 622 ing argument. A localised defect adds a total-variation increment  $\Delta TV$  that  
 623 depends only on the defect geometry ( $\Delta TV \approx 2A$  for an isolated spike of ampli-  
 624 tude  $A$ ), so the relative full-profile BV-density response is  $\Delta TV / TV_{\text{background}}$ .  
 625 On the FN 111 background ( $TV = 1.137 \mu\text{m}$ ) a 100 nm spike yields +18%;  
 626 on the FN 101 background ( $TV = 118.4 \mu\text{m}$ ) the identical spike would yield  
 627 only +0.17%, below the within-run repeatability. Detectability of a given abso-  
 628 lute defect in the full-profile descriptor therefore scales inversely with the back-  
 629 ground gradient content, which quantifies the reviewer-relevant observation that  
 630 the FN 101 defect-to-background ratio is roughly 500 times less favourable than  
 631 typical smooth-surface applications. Two consequences follow for industrial in-  
 632 spection. First, full-profile BV density is best suited to smooth, finished surfaces  
 633 — bearing raceways, optical components, precision seals — whose residual am-  
 634 plitudes (tens of nanometres) are comparable to the FN 111 case, and where the  
 635 synthetic archetypes map onto realistic features (spike  $\leftrightarrow$  pit or inclusion edge,  
 636 scratch  $\leftrightarrow$  handling or tooling mark, sinusoid  $\leftrightarrow$  chatter or vibration signature).

Second, for rougher backgrounds the sliding-window formulation restores local contrast: the certified Flick flat is detected on the FN 101 (Fig. 4) despite the unfavourable full-profile ratio, precisely because the window localises the comparison. Validation against physical defects of calibrated dimensions on real metallic and ceramic surfaces remains necessary before these scaling arguments can be considered established (Sec. 5.6).

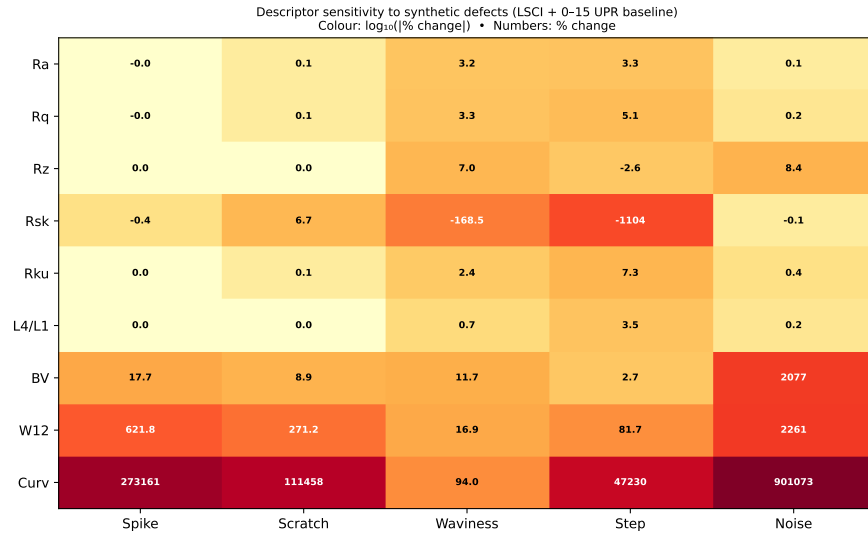


Figure 5: Sensitivity of ISO and gradient-based descriptors to five synthetic defect types injected into the FN 111 residual profile. Percentage change from the unperturbed baseline; bold labels indicate changes exceeding 10%.

Supporting analyses — descriptor correlation matrices, power spectral density comparison, Cohen’s  $d$  effect sizes, critical descriptor assessment, and  $L^p$  norm convergence — are provided in the Supplementary Materials (Figs. S13–S17). Given the number of descriptors compared ( $\sim 15$  across two artefacts), the reported differences should be interpreted as exploratory rather than confirmatory; no formal multiple-comparison correction has been applied.

**Practical considerations.** A practical suggestion for interferometric profile and roundness metrology is to retain  $R_a$ ,  $R_q$ ,  $R_z$ ,  $R_{sk}$ ,  $RONt$  and  $RONq$  as the primary engineering layer compliant with ISO 21920-2, and to consider

652 supplementing them with BV density (in angular units,  $\text{nm deg}^{-1}$ , Sec. 3.6) as  
 653 a gradient-sensitive functional-analysis complement. The  $W^{1,2}$  gradient semi-  
 654 norm is strongly correlated with BV density in the present datasets (within  
 655 the FN 111: Pearson  $r = 0.95$ ,  $n = 5$  sweeps; within the FN 101:  $r = 0.93$ ,  
 656  $n = 5$ ). The pooled  $r = 0.94$  (95% CI  $[0.72, 0.99]$ ,  $n = 10$ ) is reported for de-  
 657 scriptive purposes only — the sweeps are not independent observations and this  
 658 pooled correlation should be interpreted with caution. This strong correlation  
 659 indicates that, as *repeatability metrics across sweeps*, BV density and the  $W^{1,2}$   
 660 seminorm carry largely redundant information, and routine reporting of both  
 661 is not warranted. They are strongly correlated but not mathematically equiv-  
 662 alent: BV integrates the first power of the local slope,  $TV = \int |p'| ds$ , whereas  
 663  $W^{1,2}$  integrates its square,  $\int |p'|^2 ds$ , so the two coincide only for profiles whose  
 664 slope magnitude is nearly constant and diverge whenever the slope distribution  
 665 is uneven. This is exactly what the defect responses show: for the 100 nm  
 666 synthetic spike the  $W^{1,2}$  seminorm responded with +622% versus +18% for  
 667 BV density, because the quadratic weighting amplifies a single steep event far  
 668 more strongly than the linear total-variation measure. A practical reading of  
 669 the descriptor space in these data is therefore: (i)  $R_a/R_q$  (equivalently  $L^1/L^2$ )  
 670 carry amplitude information; (ii)  $R_{sk}$  carries an independent asymmetry axis;  
 671 (iii) BV density carries the localised-gradient axis and is recommended as the  
 672 primary gradient descriptor; (iv) the  $W^{1,2}$  seminorm adds value mainly as a  
 673 high-sensitivity alarm for isolated steep features rather than as an independent  
 674 information axis; and (v) the  $L^4/L^1$  peakiness ratio provides a supplementary  
 675 distribution-shape axis with a direct physical interpretation as the amplification  
 676 of dominant profile excursions relative to  $R_a$ . These additional descriptors are  
 677 computed from the same residual profile without additional hardware or changes  
 678 to the EFM pipeline. The cross-instrument validation suggests that gradient-  
 679 based descriptors are applicable to any 1D profile, whether from interferometry  
 680 or tactile roundness measurement.

681 **Practical value and the known-feature caveat.** It must be acknowl-  
 682 edged that the Flick flat is a certified, deliberately manufactured feature whose

683 presence and position were known in advance: Fig. 4 is an existence proof under  
684 ground-truth conditions, not a blind detection study. The practical argument  
685 for a gradient-sensitive layer is therefore not that it revealed an unknown defect  
686 here. It is that it converts what an operator might notice in a visual polar-  
687 plot inspection into a quantitative, thresholdable, and automatable indicator  
688 computed from data already acquired for ISO reporting — with no operator  
689 in the loop, a defined angular position estimate, and a repeatability statement.  
690 Whether this indicator reveals defects that routine quality-control reporting  
691 would otherwise miss has not been demonstrated; a blind-detection protocol on  
692 artefacts with independently characterised defects is outlined in Sec. 5.6 as the  
693 decisive next test of practical value.

694 **Planned versus exploratory comparisons.** The synthetic-defect re-  
695 sponses (Supplementary Table S2, Fig. 5) and the Flick flat detection (Fig. 4)  
696 were planned comparisons motivated by the study’s objective of assessing gradient-  
697 based descriptor sensitivity. The correlation matrices, power spectral density  
698 comparison, Cohen’s  $d$  effect sizes,  $L^p$  norm convergence analysis, and the cross-  
699 descriptor stability comparison (Supplementary Figs. S13–S17) are exploratory  
700 and are intended to characterise the descriptor space rather than to test specific  
701 hypotheses. This distinction should be borne in mind when interpreting the  
702 reported associations. The present study is a proof-of-concept demonstration;  
703 generalisability to other artefact types, materials, and measurement instruments  
704 remains to be established.

#### 705 5.4. Limitations

706 Several limitations should be acknowledged. First, the uncertainty analysis  
707 combines within-run Type A repeatability from five sweeps, the indicative type-  
708 B budget of Sec. 3.8 (including the measured static-session drift bound), and  
709 a two-session between-run comparison (Sec. 3.9) that shows agreement within  
710 combined uncertainties for identical acquisition parameters; a full multi-session  
711 reproducibility campaign with multiple independent setups and thermal cycles  
712 remains to be performed. The small  $n = 5$  limits statistical power (relative



713 differences below  $\sim 30\%$  cannot be resolved for the FN 111); the reported  
714 expanded uncertainties should be interpreted as lower bounds, and the form-  
715 removal systematic (bounded at  $\leq 41$  nm) is carried outside the quadrature  
716 combination.

717 Second, the validation is restricted to two artefacts from a single manufac-  
718 turer (JENOPTIK), both of  $\sim 30$  mm diameter. The two artefacts do differ  
719 substantially in material ( $\text{Al}_2\text{O}_3$  ceramic versus steel) and in residual amplitude  
720 (two orders of magnitude in  $R_a$ ), which provides some breadth, but generalis-  
721 ability to other diameters, materials such as optical glass or tungsten carbide,  
722 and other surface finishes has not been tested. Third, the synthetic-defect anal-  
723 ysis uses idealised mathematical defects (isolated spike, uniform scratch, pure  
724 sinusoid) injected into the FN 111 ceramic profile only; the response to real  
725 manufacturing defects with complex morphologies may differ, and the scaling  
726 analysis of Sec. 5.3 shows quantitatively that full-profile detection sensitivity  
727 degrades in proportion to background gradient content (the FN 101 ratio is  
728  $\sim 500$  times less favourable). Fourth, BV density and the  $W^{1,2}$  seminorm carry  
729 intrinsic sampling-interval dependence; the angular-unit convention of Sec. 3.6  
730 standardises reporting, but no inter-laboratory evidence yet exists that the con-  
731 vention transfers across instruments. Fifth, the Flick flat result (Supplemen-  
732 tary Fig. S18) is demonstrated on a single artefact with a known, certified  
733 macroscopic feature; blind-detection performance and performance for subtler,  
734 micrometre-scale defects have not been characterised.

### 735 5.5. Metrological Implications

736 The results carry several implications for profile and roundness metrology  
737 practice. The finding that BV density responds to localised gradients that am-  
738 plitude averaging may not capture suggests — subject to further validation —  
739 a potential role for gradient-based descriptors in applications where isolated  
740 defects are of concern; possible examples include quality control of bearing sur-  
741 faces, optical components, or precision seals. Because these descriptors are  
742 computed from the same residual profile used for ISO 21920-2 parameters, their

743 adoption does not require changes to existing measurement acquisition proto-  
744 cols.

745 The ISO 21920-2 framework (with ISO 4287 as its predecessor) remains the  
746 appropriate baseline for surface texture specification and conformity assessment.  
747 The gradient-based descriptors discussed here are proposed as optional supple-  
748 mentary indicators, not as replacements for standardised parameters. Their  
749 metrological utility would be strengthened by inter-laboratory comparisons, by  
750 evaluation on a wider range of artefact types, and by adoption of the angular-  
751 unit reporting convention of Sec. 3.6. Until such steps are taken, BV density and  
752 related descriptors should be regarded as research-grade tools whose diagnostic  
753 value has been observed under controlled laboratory conditions on two specific  
754 artefacts.

## 755 5.6. Future Work

756 The present study suggests several directions for further investigation. (i) A  
757 multi-session between-run reproducibility campaign with independent setups,  
758 distinct thermal cycles, and physical re-centring of the artefact would extend  
759 the two-session comparison of Sec. 3.9 and would supply the input distribu-  
760 tions needed to upgrade the indicative type-B budget of Sec. 3.8 to a complete  
761 GUM-compliant evaluation. (ii) Extending the validation to artefacts with dif-  
762 ferent diameters, materials (e.g., optical glass, tungsten carbide), and surface  
763 finishes would test the generalisability of the findings beyond the two  $\sim 30$  mm  
764 JENOPTIK artefacts studied here. (iii) A systematic characterisation of BV  
765 density response to real manufacturing defects (scratches, pits, inclusions) of  
766 calibrated dimensions, culminating in a blind-detection protocol in which the  
767 analyst does not know defect presence or position, would bridge the gap be-  
768 tween the present synthetic-defect analysis and industrial inspection and would  
769 provide the decisive test of practical value. (iv) An inter-laboratory compari-  
770 son using a common artefact set and a standardised processing pipeline would  
771 provide the reproducibility data needed to assess whether BV density could  
772 eventually be considered for standardisation. (v) Empirical verification of the

angular-unit convention of Sec. 3.6 across instruments with different angular resolutions would support the potential inclusion of gradient descriptors in reference software for roundness evaluation. (vi) A systematic study of the speed–accuracy trade-off (rotation speed, frame rate, fringe contrast) would establish the shortest acquisition compatible with the reported uncertainty, a prerequisite for industrial adoption.

## Data Availability

The analysis code, processed results, and all figures are available at <https://github.com/dawidkucharski/banach-profile-metrology> (release v1.1.0). The raw interferometric PNG archives (~32 GB), together with the two additional independent FN 111 sessions used for the between-run comparison of Sec. 3.9 and the static stability recording used for the drift bound in Sec. 3.8 (~7 GB each), are available from the corresponding author upon reasonable request.

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